Chiral Thiophosphoramide and Thioamide Ligands in Catalytic Asymmetric Carbon–Carbon Bond-Formation Reactions

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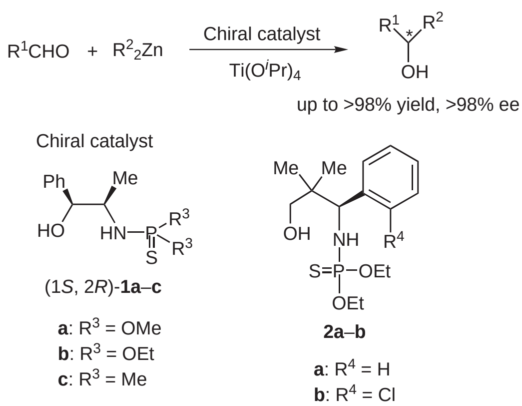
Abstract: Metal complexes with organosulfur ligands exist widely in nature. The high polarizability of the electrons on the sulfur atoms is responsible for a variety of structures and the reactivity of metal complexes with sulfur-containing ligands. Organosulfur-containing chiral ligands have various applications in catalytic asymmetric reactions. Our group has been interested in the applications of a series of thiophosphoramide and thioamide ligands in asymmetric carbon–carbon bond-formation reactions. In this account, we describe the synthesis of chiral thiophosphoramide and thioamide ligands and their applications in catalytic asymmetric carbon–carbon bond formation.

Key words: organosulfur-containing ligands, thiophosphoramide, thioamide, asymmetric carbon–carbon bond formation

1 Introduction

Some of the most ubiquitous and diverse metal-containing structures in biology are iron–sulfur clusters.[1] Since the early 1960s iron–sulfur complexes have been known for their role in electron transfer, most notably in ferredoxins, mitochondrial electron transport chains, and photosynthesis.[2–4]The high affinity of metals for sulfur is clearly demonstrated by the enormous variety of metal–sulfide minerals that exist in nature.[5] Qualitatively this might be understood using Pearson's concept of hard and soft acids and bases.[6] However, it is better described by the covalence of the metal–sulfur bond, particularly with the 'soft' metals. In contrast to oxygen and nitrogen as donor atoms, sulfur has low lying, unoccupied 3*d*-orbitals available. This influence might be overestimated by molecular orbital calculations reported in the literature.[7] However, it has to be assumed that the high polarizability of the electrons on the sulfur atoms is responsible for the variety of structures and the reactivity of the metal complexes with sulfur-containing ligands. These π-accepting ligands will decrease the electron density on the metal and thus allow for an increased electron density in the metal–sulfur bond.

The coordinating ability of sulfur-containing ligands (i.e. the stability constants of the complexes) increases with dipole moment in the order $H_2S<RSH<R_2S$ and the stability of the complexes decreases in the order $S^{2-}>RS^{-}>R_2S$.[8] The polarizability and the number of lone electron pairs in the sulfur-containing ligands are the dominant factors in coordinating ability and the stability of the metal complexes. In contrast to the large number of complexes using water as a ligand, very few examples have been reported containing hydrogen sulfide as a ligand. This could be due to the fact that the primary product can easily react to yield sulfhydryl or multinuclear complexes with bridging sulfur ligands.[9] The first stable hydrogen sulfide complexes were reported in 1976.[10] Other sulfur-containing ligands, such as thioethers,[11] oxides of sulfur,[12] thioureas,[13] dithiobiurets,[14] dithiolenes,[15] phosphine sulfides,[16] amino-substituted phosphine sulfides,[17] sulfur-containing oxazolines[18] and polysulfides[19] are widely utilized in organic reactions. Among the various sulfur-containing ligands of interest today, using chiral thiophosphoramides as ligands in catalytic asymmetric carbon–carbon bond-formation reactions should be considered. Thiophosphoramidates were first utilized as chiral ligands by Soai[20] on the basis of his report on the enantioselective synthesis of chiral dialkyl phosphoramides by the alkylation of phosphinoyl-imines.[21] Since then, more chiral thiophosphoramidate ligands have been synthesized and utilized as



Scheme 1

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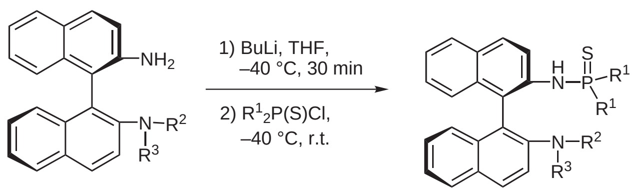
catalysts for the addition of dialkylzincs to aldehydes in the presence of titanium(IV) isopropoxide (Scheme 1).[22] Additionally, our group has synthesized a series of thiophosphoramides in order to examine their use in asymmetric reactions. Herein, we summarize our efforts in the synthesis and applications of this series of thiophosphoramide and thioamide ligands in asymmetric carbon–carbon bond-formation reactions.

2 Synthesis of Chiral Organosulfur-Containing Ligands

The chiral binaphthylthiophosphoramide ligands **L1**–**L7** were synthesized from (*R*)-1,1′-binaphthyl-2,2′-diamine (BIMAM),[23] (*R*)-N^2-ethyl-1,1′-binaphthalenyl-2,2′-diamine (**3**) and (*R*)-$N^{2\prime}$,$N^{2\prime}$-dimethyl-1,1′-binaphthalenyl-2,2′-diamine (**4**) in very high yields via the lithiation of relevant diamines with butyllithium and then phosphorylation with dialkyl-, dialkoxyl- or diarylthiophosphoryl chloride (Scheme 2). Synthesis of the C_2-symmetric chiral phosphoramide from BINAM was unsuccessful. However, similar phosphoramide ligands **L8**–**L11** synthesized from (1*R*,2*R*)-(–)-1,2-diaminocyclohexane (**5**) and (1*S*,2*S*)-(–)-1,2-diphenylethylenediamine[24] (**6**) were obtained in good yields in the presence of diisopropylethylamine or triethylamine in dichloromethane or butyllithium in tetrahydrofuran, respectively (Scheme 3). Chiral C_1-symmetric diaminothiophosphoramide ligands **L12**–**L15** were obtained from (1*R*,2*R*)-(–)-1,2-diaminocyclohexane (**5**) and (1*R*,2*R*)-(+)-1,2-diphenylethylenediamine (**6**) through similar methods as those described above (Scheme 4). N-Substituted dimethylthiophosphoramide ligand **L16** was prepared from (1*R*,2*R*)-(–)-*N*-(2-aminocyclohexyl)acetamide (**9**), after reduction of (1*R*,2*R*)-(–)-*N*-(2-aminocyclohexyl)-acetamide (**8**) using lithium aluminium hydride (LAH) in tetrahydrofuran (Scheme 5).[25]



	R2	R3
(*R*)-BINAM	H	H
(*R*)-**3**	H	Et
(*R*)-**4**	Me	Me

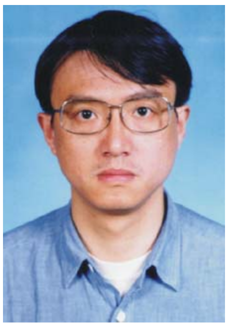
	R1	R2	R3	Yield
L1	Me	H	H	95%
L2	Me	H	Et	90%
L3	Ph	H	H	92%
L4	Ph	H	Et	93%
L5	Ph	Me	Me	51%
L6	OEt	H	H	88%
L7	OEt	H	Et	97%

Scheme 2

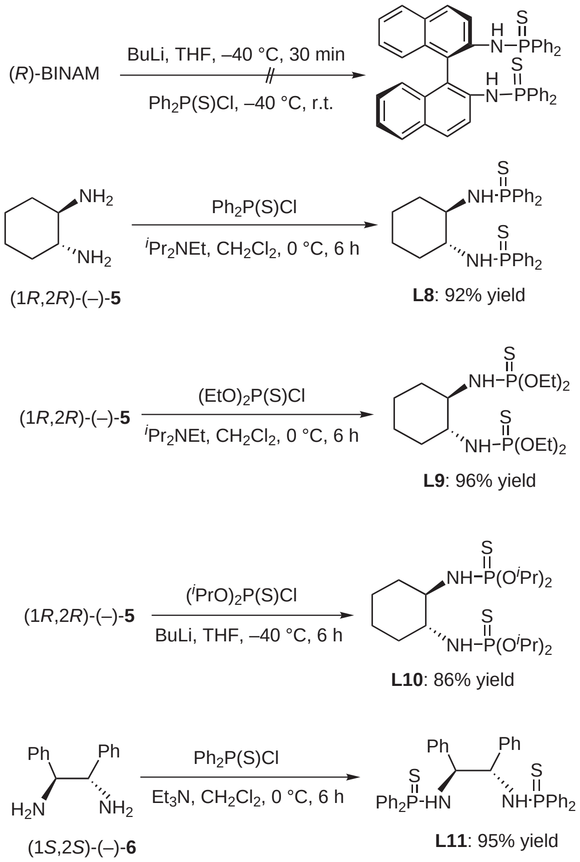
The unsubstituted amino group in **L12** was easily modified to give a series of chiral thiophosphoramide ligands through one-step reactions. Chiral imino-thiophosphoramide ligands **L17**–**L29** were readily synthesized in moderate to good yields by reaction of **L12** with various arylaldehydes in ethanol either at room temperature or under reflux (for sterically bulky arylaldehydes) for four to five hours (Scheme 6).

Biographical Sketches

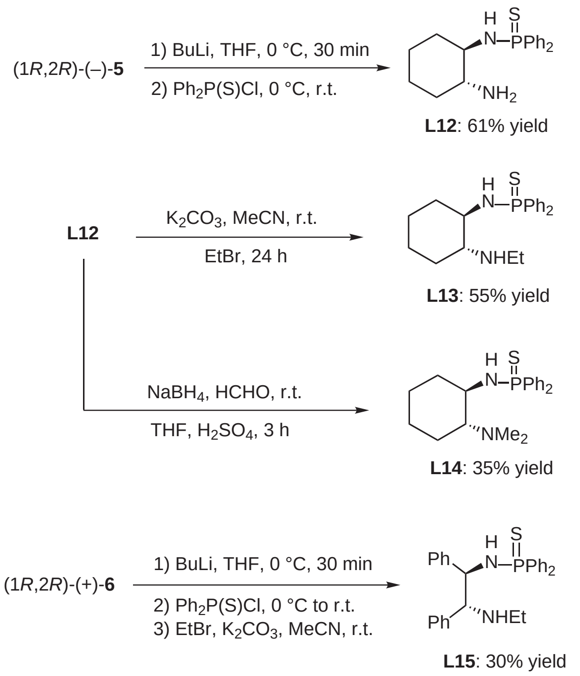
Wen Zhang was born in 1978 in Hubei, China and graduated from Wuhan University in 2001. She is currently in the final stage of her PhD degree in the State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry (SIOC) under the supervision of Professor Min Shi. Her research is concerned with the synthesis of chiral sulfur and phosphine ligands and their applications in the asymmetric addition of diethylzinc to enones and imines.



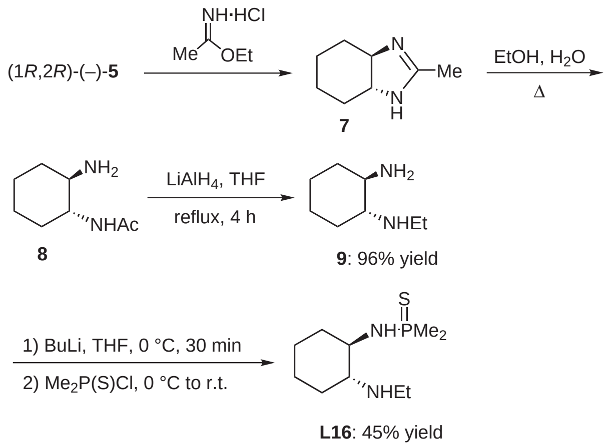
Dr. Min Shi was born in 1963 in Shanghai, China. He received his BS in 1984 from the Institute of Chemical Engineering of East China and his PhD in 1991 from Osaka University, Japan. His first postdoctoral research experience was with Prof. Kenneth M. Nicholas at the University of Oklahoma (1995–1996). He then worked as an ERATO Researcher in the Japan Science and Technology Corporation (JST) from 1996–1998. He is currently a group leader in the State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry (SIOC).



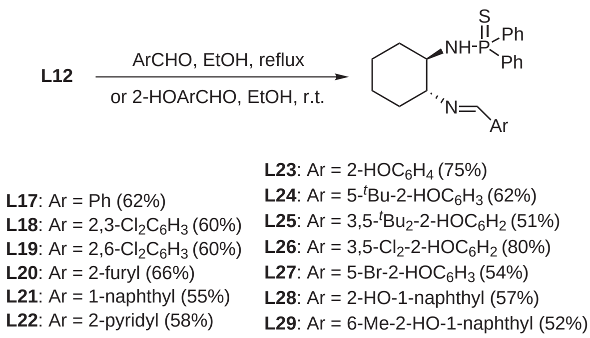
Scheme 3



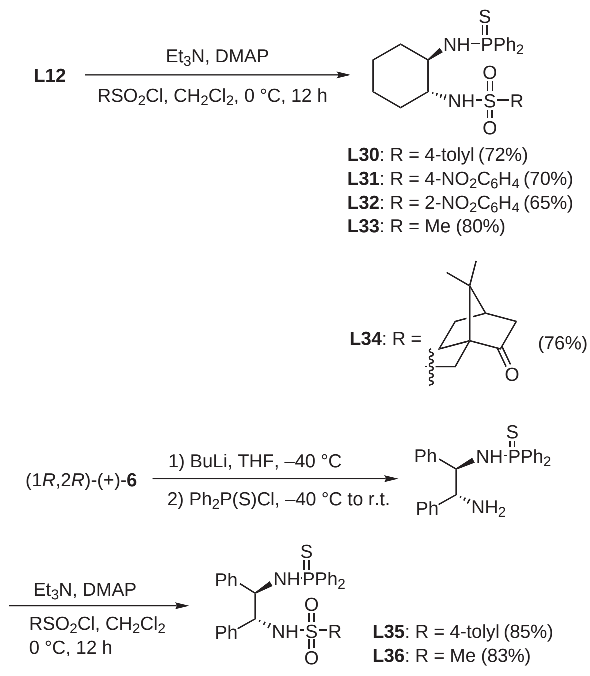
Scheme 4



Scheme 5



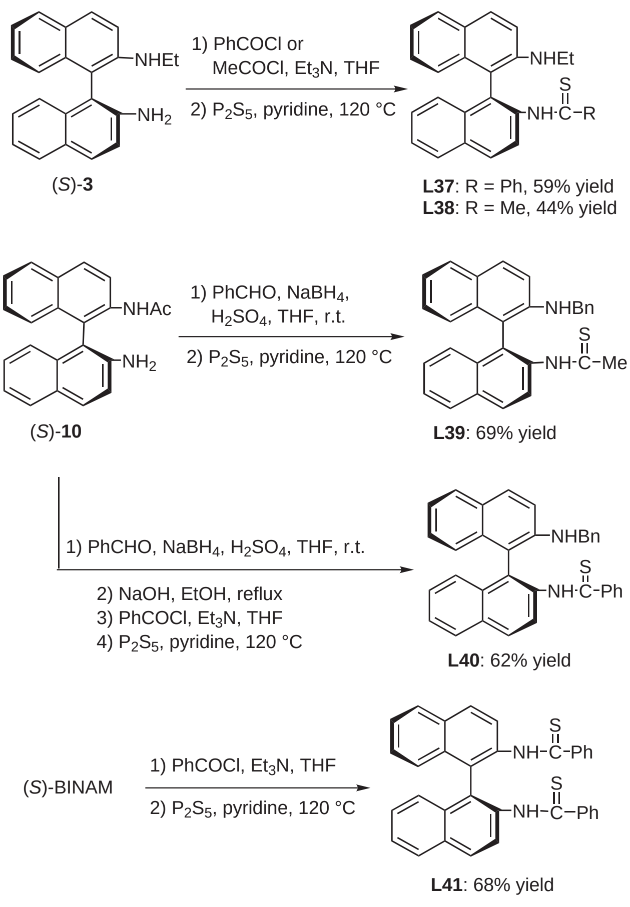
Scheme 6



Scheme 7

Moreover, chiral sulfonamide-thiophosphoramide ligands **L30**–**L36** were readily synthesized from the reaction of **L12** and (1*R*,2*R*)-(+)-**6** with various sulfonyl chlorides in the presence of triethylamine and a catalytic amount of 4-(*N*,*N*-dimethylamino)pyridine. Moderate to good yields were obtained in 12 hours in dichloromethane at 0 °C (Scheme 7).[26]

The axially chiral thioamide ligands **L37**–**L41** were readily synthesized in moderate yields from the reaction of (*S*)-BINAM with arylaldehydes or acyl chlorides and phosphorus pentasulfide (P_2S_5). This presumably coordinates to the metal center by interaction with either the sulfur and nitrogen atom or with two sulfur atoms in the catalytic asymmetric carbon–carbon bond formation (Scheme 8).[27]



Scheme 8

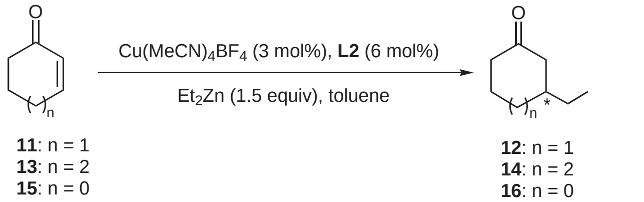
3 Applications in Carbon–Carbon Bond-Formation Reactions

3.1 Conjugate Addition of Organozinc Reagents to Enones

3.1.1 Utilization of Thiophosphoramide and Thioamide Ligands in the Addition of Organozinc Reagents to Enones

The conjugate addition of various organometallic reagents to α,β-unsaturated carbonyl compounds is an important process in catalytic enantioselective carbon–carbon bond-forming reactions.[28] A critical aspect of these investigations is the utilization of such asymmetric methods in the total synthesis of biologically active molecules.[29] We have explored the catalytic enantioselective conjugate addition of dialkylzinc or diphenylzinc to enones with our organosulfur-containing ligands. We found that these organosulfur-containing ligands provide an efficient and highly enantioselective functionalization protocol of not only six- and seven-membered cyclic enones, but also cyclopentenone, acyclic aromatic enones and aliphatic enones.[30] Chiral binaphthylthiophosphoramide ligand **L2** proved to be the best ligand in the asymmetric addition reaction of a diethylzinc reagent to enones. This gave 75–93% yields with 97–98% ee for five-, six- and seven-membered cyclic enones using $Cu(MeCN)_4BF_4$ as a precursor in toluene (Table 1).[30a]

Table 1 Asymmetric 1,4-Addition Reaction of Enones with Diethylzinc Catalyzed by $Cu(MeCN)_4BF_4$ and Chiral Ligand **L2** under Optimal Conditions



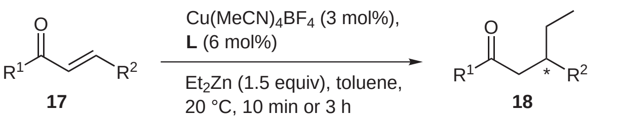
Entry	n	Temp (°C)	Time (h)	Yield (%)[a]	ee (%)[b]	Config[c]
1	1	0	0.5	95	97	*R*
2	2	20	0.5	93	97	*R*
3	0	0	0.5	75	98	*R*

[a] Isolated yield.
[b] Determined by GC.
[c] Determined by the sign of the specific rotation.

Chiral ligands **L2** and **L4** were also used in the addition reaction of diethylzinc to acyclic aromatic and aliphatic enones (Table 2). The enantioselectivities and yields were very high for acyclic aromatic enones and these reactions were complete within ten minutes. In the case of acyclic aliphatic enones, the yields were 82–92% with 80–93% ee within three hours.

Organozinc compounds represent ideal reagents for copper-catalyzed enantioselective Michael addition reactions because of their low reactivity towards the substrate in the absence of a copper catalyst. Diethylzinc is nearly always the alkylating agent used in asymmetric conjugate addition reactions of dialkylzinc reagents to enones catalyzed by copper salts. Dimethylzinc and diphenylzinc have seldom been used, as they are much less reactive and generally need longer reaction times for the asymmetric Michael addition reaction.[31] In order to clarify the scope and limitations of organozinc reagents in our copper(I)–thiophosphoramide ligand system. We examined the asymmetric conjugate addition reaction of dimethylzinc and diphenylzinc to 2-cyclohexen-1-one (**11**) in the presence of **L2** or **L4** under our optimized conditions. The results are shown in Table 3. The relatively less reactive dimethylzinc and diphenylzinc can be employed in the asymmetric conjugate addition of **11** efficiently with appreciable asymmetric induction in our catalytic system. Using **L2** or **L4** as the chiral ligand, transformation with

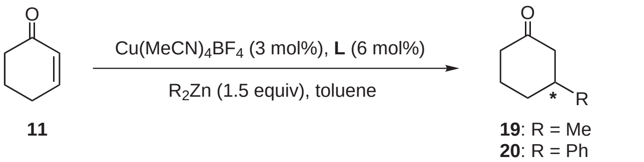
Table 2 Asymmetric 1,4-Addition Reaction of Acyclic Enones with Diethylzinc Catalyzed by $Cu(MeCN)_4BF_4$ and Chiral Ligands **L2** and **L4** under Optimal Conditions



Entry	R[1]	R[2]	Product	Ligand	Yield (%)[a]	ee (%)[b]
1	Ph	Ph	**18a**	**L4**	98	96
2	1-naphthyl	Ph	**18b**	**L4**	98	96
3	Ph	*p*-BrC_6H_4	**18c**	**L4**	99	96
4	Ph	*p*-$MeOC_6H_4$	**18d**	**L4**	98	96
5	*p*-BrC_6H_4	Ph	**18e**	**L4**	97	95
6	*p*-$MeOC_6H_4$	Ph	**18f**	**L4**	97	97
7[c]	Me	*n*-Bu	**18g**	**L2**	89	93
8[c]	Me	*i*-Pr	**18h**	**L2**	82	90
9[c]	*i*-Bu	*n*-Bu	**18i**	**L2**	92	85
10[c]	*i*-Bu	*n*-Bu	**18j**	**L2**	92	80

[a] Isolated yield.
[b] Determined by chiral GC.
[c] Diethylzinc was first added into the reaction mixture before the addition of the corresponding enones.

Table 3 Asymmetric 1,4-Addition Reaction of 2-Cyclohexen-2-one with R_2Zn Catalyzed by $Cu(MeCN)_4BF_4$ and Chiral Ligands

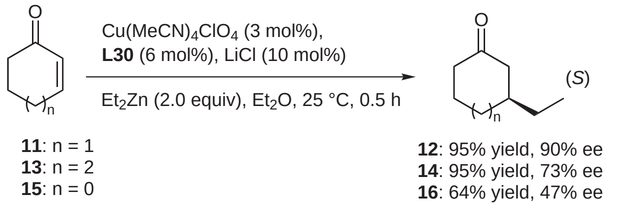


Entry	Ligand	R	Temp (°C)	Time (h)	Yield (%)[a]	ee (%)[b]	Config[c]
1	**L2**	Me	0	6	77	85	*R*
2	**L2**	Me	20	4	71	87	*R*
3	**L4**	Me	0	6	73	79	*R*
4	**L4**	Me	20	4	70	85	*R*
5	**L2**	Ph	0	8	97	89	*R*
6	**L4**	Ph	0	8	98	77	*R*

[a] Isolated yield.
[b] Determined by chiral HPLC or GC.
[c] Determined by the sign of the specific rotation.

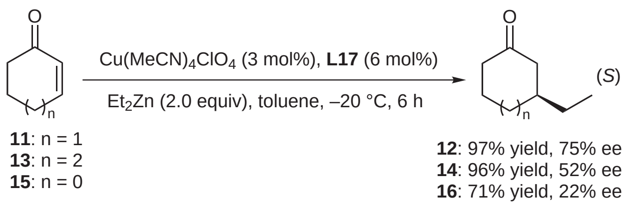
dimethylzinc gave the desired product in 70–77% yields with 79–87% ee within four to six hours (Table 3, entries 1–4). Using diphenylzinc in the presence of **L2** or **L4**, the reaction was more sluggish, but still gave the desired product in almost quantitative yields with up to 89% ee, better than results previously reported in the literature[32] (Table 3, entries 5 and 6).

Chiral sulfonamide-thiophosphoramide ligands **L30–L36** were also utilized as chiral ligands in the $Cu(MeCN)_4ClO_4$-promoted catalytic asymmetric addition of diethylzinc to cyclic enones. Using ligand **L30** and lithium chloride as an additive, up to 90% ee was realized at room temperature (25 °C) within half an hour for a six-membered cyclic enone (Scheme 9).[30b] In the absence of lithium chloride, 62% ee was achieved under identical conditions.[30b] The improvement on chiral induction with the addition of lithium chloride may be due to the fact that lithium can coordinate to the oxygen atom in the sulfonamide ligand **L30**, leading to a rigid transition state that improves enantioselectivity. However, this additive is not always beneficial when used with other phosphoramide ligands. When cyclopentenone and cycloheptenone were used, only moderate enantioselectivities were achieved, presumably due to the fact that sterically rigid or flexible cyclic enones can not join perfectly into the chiral pockets derived from these sulfonamide-thiophosphoramide ligands.



Scheme 9

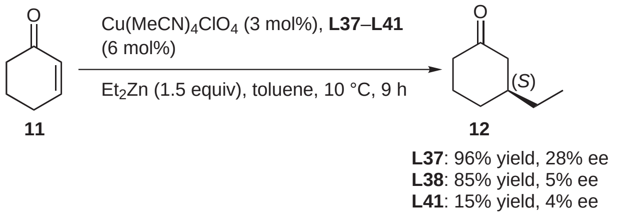
In the presence of a catalytic amount of chiral imino-thiophosphoramide ligands **L17–L29** (6 mol%) and copper(I) salt (3 mol%), the asymmetric addition of diethylzinc to α,β-unsaturated carbonyl compounds could be achieved in good yields and moderate ee (up to 75% ee using **L17** as ligand) at –20 °C in toluene (Scheme 10).[30c]



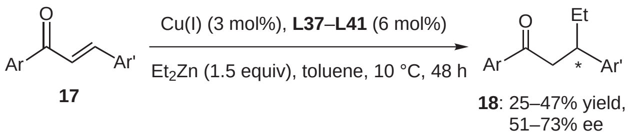
Scheme 10

The axially chiral thioamide ligands **L37**–**L41** have also been widely used in the asymmetric addition of diethylzinc to 2-cyclohexen-1-one (**11**) and various chalcone derivatives **17** to achieve moderate to good yields and ee values (Schemes 11 and 12).[30d]

It should be noted that all the chiral thiophosphoramide and thioamide ligands utilized in the addition of organozinc reagents to enones are easily available, quite stable and recoverable. They can be recovered by column



Scheme 11

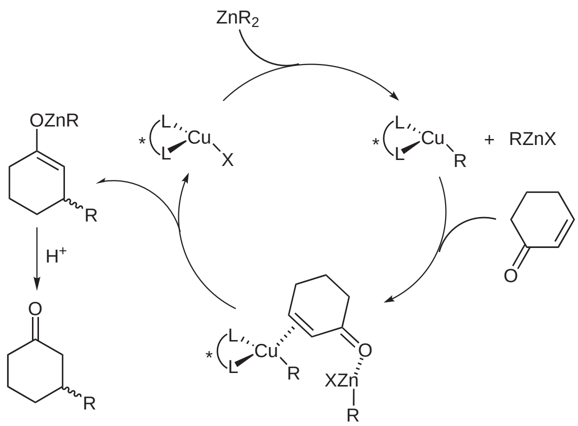


Scheme 12

chromatography and reused in similar addition reactions without loss of yield and enantioselectivity.

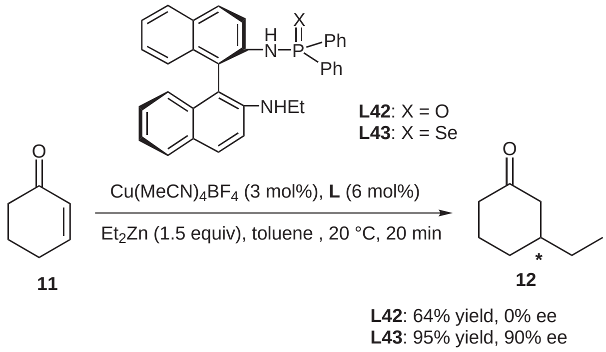
3.1.2 Mechanistic Studies

According to previous literature,[33] the proposed pathway for the catalytic asymmetric conjugate addition is as shown in Scheme 13. The mechanism of the reaction calls for the transfer of an alkyl group from zinc to a chiral copper complex, which is subsequently capable of delivering the alkyl group to the enone with enantioselectivity.



Scheme 13

It is well known that sulfur atoms can strongly coordinate to late transition metals.[34] For example, copper(I) compounds have a greater affinity for soft ligands (olefins, sulfur, phosphorus and selenium atoms).[35] In order to further verify this speculation, we synthesized diphenylphosphoramide ligand **L42** and diphenylselenophosphoramide ligand **L43** from (*R*)-BINAM using the same method as described above and applied them in the asymmetric conjugate addition reaction of diethylzinc to 2-cyclohexen-1-one (**11**) (Scheme 14). It was found that the sulfur atom bonded to the phosphorus atom was crucial in order for the catalytic asymmetric reaction to be effective because the corresponding axially chiral diphenylphosphoramide ligand **L42** showed no enantioselectivity in this conjugate addition reaction under the same conditions (Scheme 14).[36] This may be due to the fact that the oxygen atom is a harder ligand than the sulfur atom and can not smoothly coordinate to a softer metal center, such as the copper(I) center, according to the soft–hard acid and base theory.[37] On the other hand, according to this theory, sulfur and selenium atoms are thought to have great affinities for copper(I) compounds. We therefore believe that chiral ligand **L43** should have similar catalytic abilities to ligand **L4**. As can be seen from Scheme 14, when using **L43** as a chiral ligand under these optimal reaction conditions, 95% yield and 90% ee were also obtained within 20 minutes at room temperature (20 °C), very similar results to those seen in the conjugate addition of diethylzinc to enones using **L4** as the chiral ligand.

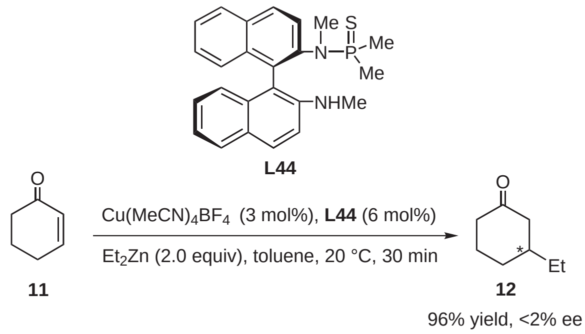


Scheme 14

In order to better understand the coordination of the thiophosphoramide ligands to copper(I), the ^{31}P NMR spectra measurements of ligands **L2**, **L4** and **L7** were carried out in the absence and presence of the copper(I) salt. The apparent changes in chemical shift were observed by comparing the corresponding ^{31}P NMR spectra of chiral ligands **L2**, **L4** and **L7** with those run in the presence of $Cu(MeCN)_4BF_4$ (1:1 mixture) in deutero-chloroform at room temperature. In the absence of $Cu(MeCN)_4BF_4$, the phosphorus signal of **L2**, **L4** and **L7** appeared at 56.68 ppm, 53.28 ppm and 66.57 ppm, respectively. In the presence of $Cu(MeCN)_4BF_4$, there was a downfield shift of the phosphorus signal to 62.37 ppm and 55.52 ppm for **L2** and **L4**, respectively and an upfield shift to 64.99 ppm for **L7**. However, for **L42**, where the phosphorus atom is connected to oxygen, no chemical shift was observed under similar conditions. These results suggest that the sulfur atom on the thiophosphoramide is probably coordinated to the copper(I) metal center.

Besides the sulfur atom, another precoordinative atom in our chiral thiophosphoramide ligand system is the nitrogen atom in $ArNR^2R^3$. ^{13}C NMR studies of the **L4**/$Cu(MeCN)_4BF_4$ complex (1:1 mixture) in deutero-chloroform at room temperature were carried out. In the absence of $Cu(MeCN)_4BF_4$, the carbon signals of the two carbons in the ethyl group of **L4** appeared at 38.31 ppm and 15.00 ppm, but the two corresponding carbon signals appeared at 40.92 ppm and 14.88 ppm in the presence of $Cu(MeCN)_4BF_4$. Another new carbon signal was also observed at 1.88 ppm, which was attributed to the carbon of the methyl group in acetonitrile. These results suggest that $Cu(MeCN)_4BF_4$ could potentially be coordinated by both S and N atoms in the thiophosphoramide ligands, although at the present stage, we do not have a crystal structure of this chiral copper(I)–ligand complex.

Noyori and Kitamura have reported that using a mixture of copper cyanide and *N*-benzylbenzenesulfonamide as additive catalyzed the conjugate addition of dialkylzincs or diarylzincs to enones to give the desired products in nearly quantitative yields with high efficiency.[38] They found that N-monosubstituted sulfonamides worked much better than N-unsubstituted compounds, but a bulky group on the nitrogen atom considerably lowered the reaction rate. Moreover, *N*,*N*-dibenzylbenzenesulfonamide was totally ineffective. In their proposed catalytic cycle, they suggested that a bimetallic catalytic cycle accelerated the reaction and the acidic proton in *N*-benzylbenzenesulfonamide played a significant role in their catalytic system. In order to verify the postulation, we synthesized the ligand **L44** from N^2,$N^{2\prime}$-dimethyl[1,1′]binaphthalenyl-2,2′-diamine[23] using the same method as described above and applied it in the same addition reaction of diethylzinc to 2-cyclohexen-1-one (**11**) (Scheme 15). Under these optimal reaction conditions, chiral ligand **L44** showed no asymmetric induction effect for the addition of diethylzinc to cyclohexenone (<2% ee), although the reaction was complete after 30 minutes at room temperature.



Scheme 15

From this remarkable change in enantioselectivity, caused by modification of the ligand structure, we concluded that the acidic proton of thiophosphoramide in our ligand system also played a significant role in the catalytic reaction process, very similar to previously reported work.[32,38] In addition, we also confirmed that in the ^{1}H NMR spectrum of **L2**, the acidic proton in the NH-P(*S*)Me_2 moiety at 4.61 ppm disappeared in the presence of diethylzinc and its ^{31}P NMR spectrum with copper(I) clearly indicated a downfield chemical shift from 56.68 ppm to 61.56 ppm in the presence of diethylzinc. These results suggest that the active species shown in Figure 1 might exist in the reaction system.

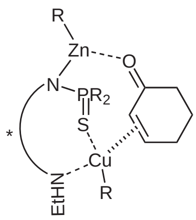


Figure 1

The study of nonlinear effects in asymmetric catalysis has become an important mechanistic tool.[39] The enantiomeric excess observed in the product can be linearly proportional to the enantiomeric excess of the catalyst or have a non-linear relationship which can be either positive (asymmetric amplification) or negative (asymmetric depletion) according to Kagan's models.[40] We next examined the relationship between the enantiomeric excess of the product and that of the ligand. A clear linear effect was observed in our reaction system. Such a linear correlation between product ee and ligand ee indicates that the active species was a monomeric copper(I) complex bearing a single chiral ligand [copper(I)–ligand, 1:1]. These results suggest that our thiophosphoramide ligand system in the copper(I)-promoted conjugate addition reaction was a single S,N-bidentate ligand to copper(I) combined with a bimetallic catalytic cycle.

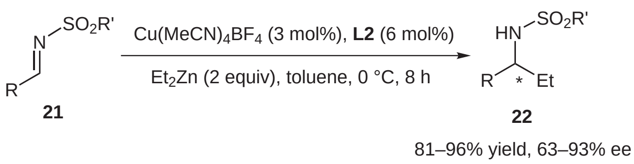
Mechanistic studies on the chiral sulfonamide-thiophosphoramide ligands **L30**–**L36** to the copper center using ^{1}H NMR, ^{31}P NMR and ^{13}C NMR spectroscopic experiments also indicated this type of ligand to be an S,O-bidentate ligand. In addition, the linear relationship between ligand ee and product ee revealed that the active species was a monomeric copper(I) complex bearing a single ligand, similar to the binaphthylthiophosphoramide ligands in our reaction system.

3.2 Addition Reaction of Diethylzinc to Imines

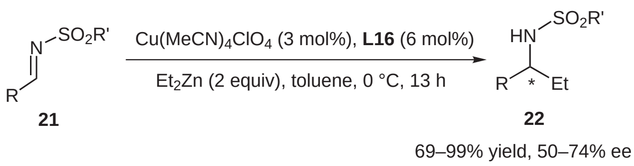
3.2.1 Addition Reaction of Diethylzinc to *N*-Sulfonylimines

The efficient and catalytic asymmetric preparation of amines is one of the most promising methodologies in homogeneous catalysis.[41] Enantioselective addition of organometallic reagents to the C=N of imines is a convenient route to obtaining optically active amines.[42] This approach maintains good enantioselectivity and has been used in chiral amine ligand-catalyzed asymmetric addition of alkyllithium,[43] chiral aminoalcohol ligands,[44] copper-amidophosphines,[45] zirconium-peptide-based chiral ligands,[46] chiral allylpalladium-catalyzed allylation by allylstannanes[47] and rhodium-monophosphine-catalyzed arylation by arylstannanes.[48]

We examined this asymmetric addition of *N*-sulfonylimines with diethylzinc in toluene under various reaction conditions using copper sources and binaphthylthiophosphoramide ligands **L1**–**L7**[49] or C_1-symmetric diaminothiophosphoramides **L12**–**L15**.[50] These are easily available, stable, recoverable and reusable chiral ligands used in asymmetric addition reactions. These ligands and reaction conditions have been carefully screened and examined to develop the best reaction conditions. Under these optimized reaction conditions, good yields and up to 93% ee with ligand **L2** (Scheme 16) and 74% ee with ligand **L16** (Scheme 17) were obtained in toluene at 0 °C. The best results are shown in Table 4, these use **L2** as the chiral ligand in the asymmetric addition reaction of diethylzinc to various aryl and aliphatic substituted *N*-sulfonylimines. As can be seen from Table 4, high yields and good enantioselectivities could be achieved for various aromatic *N*-sulfonylimines **21** having either electron-donating or electron-withdrawing groups on the benzene ring (Table 4, entries 1–12). These results are comparable with the best literature values hitherto reported for the asymmetric addition of diethylzinc to *N*-sulfonylimines. Using methanesulfonylimine and 2-trimethylsilylethanesulfonylimine (SES), 96% yield (87% ee) and 94% yield (80% ee) could also be realized (Table 4, entries 13 and 14). In the case of the aliphatic sulfonylimine, only a moderate ee (63% ee) for the corresponding addition product **22o** was obtained (Table 4, entry 15).



Scheme 16



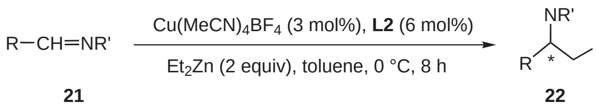
Scheme 17

3.2.2 Addition Reaction of Diethylzinc to *N*-Phosphinoylimines

The enantioselective synthesis of α-chiral amines is of primary importance because these chiral synthons are among the most commonly found and important subunits in chiral drugs.[42] In addition to the asymmetric reduction of imines, the enantioselective addition of alkylmetals to imines is a convenient route to optically active amines. Recently, the use of *N*-phosphinoylimines as activated substrates in combination with diethylzinc as the nucleophilic reagent and amino alcohol as the promoter is receiving increasing attention because the phosphoryl group can be easily removed under acidic conditions.[44,51]

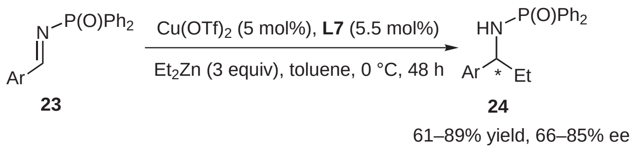
We used our binaphthylthiophosphoramide ligands **L1**–**L7** in the catalytic asymmetric addition of diethylzinc to *N*-phosphinoylimines. Using **L7** as the chiral ligand in toluene at 0 °C, 61–89% yields and up to 85% ee were obtained with various aromatic substituted *N*-phosphinoylimines (Scheme 18).[52] This method provides an alternative synthetic approach to the corresponding α-chiral amines.

Table 4 Asymmetric Addition Reactions of Diethylzinc to Various *N*-Sulfonylimines **21** in the Presence of Copper(I) (3 mol%) and Chiral Ligand **L2** (6 mol%)



Entry	R	R′	Product	Yield (%)[a]	ee (%)[b]	Config[c]
1	Ph	Ts	**22a**	93	86	*S*[c]
2	*p*-MeC_6H_4	Ts	**22b**	95	86	–[d]
3	*m*-MeC_6H_4	Ts	**22c**	93	86	–[d]
4	*p*-$MeOC_6H_4$	Ts	**22d**	81	90	–[d]
5	*p*-ClC_6H_4	Ts	**22e**	96	91	–[d]
6	*m*-ClC_6H_4	Ts	**22f**	93	86	–[d]
7	*p*-$CF_3C_6H_4$	Ts	**22g**	95	93	–[d]
8	*p*-BrC_6H_4	Ts	**22h**	91	92	–[d]
9	*m*-FC_6H_4	Ts	**22i**	94	85	–[d]
10	*p*-FC_6H_4	Ts	**22j**	96	90	–[d]
11	2-furyl	Ts	**22k**	95	82	–[d]
12	1-naphthyl	Ts	**22l**	93	88	+[d]
13	Ph	Ms	**22m**	96	87	–[d]
14	*p*-ClC_6H_4	SES	**22n**	94	80	–[d]
15	Me_2CH	Ts	**22o**	86	63	–[d]

[a] Isolated yields.
[b] Determined by chiral HPLC.
[c] The absolute configuration was assigned by comparison of the optical rotation with reported data.
[d] Sign of the optical rotation.

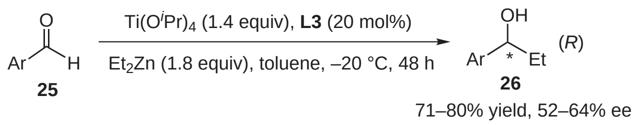


Scheme 18

3.3 Addition Reaction of Diethylzinc to Aldehydes

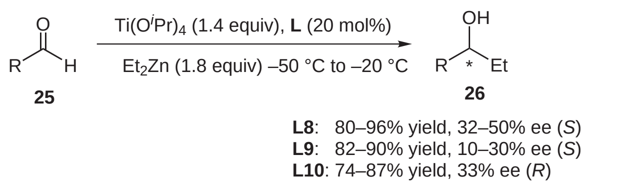
The catalytic enantioselective addition of organozinc reagents to aldehydes is a versatile method in the synthesis of optically active secondary alcohols during carbon–carbon bond-forming reactions.[53,54] Chiral zinc and titanium complexes can catalyze this reaction when used in combination with amino alcohols (N,O ligands),[55] diamines

(N,N ligands)[56] or diols (O,O ligands),[57] which act as excellent chiral ligands. Binaphthylthiophosphoramide ligands, which were successfully utilized in the addition of diethylzinc to enones or imines, were applied in the addition of diethylzinc to aldehydes (Scheme 19).[58] Using various arylaldehydes as substrates and in the presence of **L3** (20 mol%), 71–80% yields with 52–64% ee were obtained at –20 °C in toluene in 48 hours.



Scheme 19

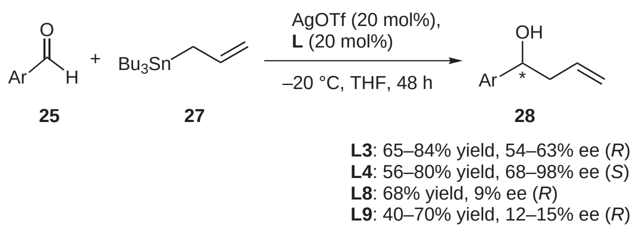
As well as the C_1-symmetric thiophosphoramide ligands, several C_2-symmetric thiophosphoramide ligands **L8**–**L10** were also successfully applied in this addition reaction (Scheme 20).[59] Good yields with moderate ee values were obtained in this Ti(O*i*-Pr)$_4$-ligand catalytic system for a variety of arylaldehydes at –50 °C → –20 °C.



Scheme 20

3.4 Asymmetric Addition of Allylic Stannane to Aldehydes

Catalytic asymmetric synthesis is a valuable method in the preparation of optically active substances, particularly in the asymmetric allylation of carbonyl compounds to give optically active secondary homoallylic alcohols.[60] The stereoselective addition of allylic organometallics to one of the two heterotopic faces of a carbonyl group has been extensively studied and the enantioselective allylation of various aldehydes is of particular importance in organic synthesis. Although significant results have been achieved using a stoichiometric amount of chiral Lewis acid,[61] there are only a few methods reported that use a catalytic process. These include using chiral (acyloxy)borane (CAB) complex/allylic silanes,[62] allylic stannanes,[63] binaphthol-derived chiral titanium complexes with allylic stannanes,[64] allyllation using a BINAP silver(I) complex[65] and chiral phosphoramides.[66]

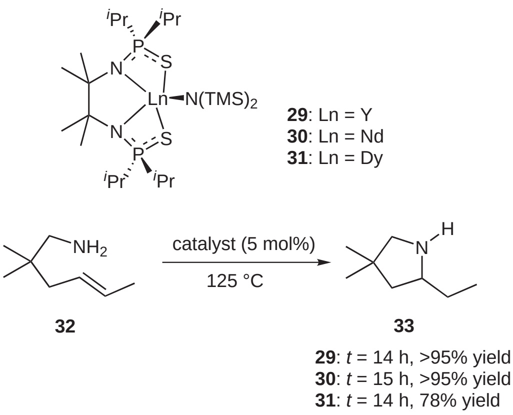


Scheme 21

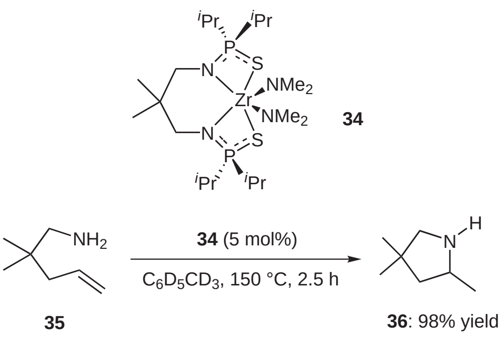
We previously reported a catalytic enantioselective allylation reaction of aldehydes with allyltributyltin promoted by silver(I) (AgOTf)-chiral thiophosphoramide **L1**–**L9**, in which a high enantioselectivity (up to 98% ee) was attained using ligand **L4** (Scheme 21).[67] This is a novel example using a chiral organosulfur-containing ligand in the catalytic asymmetric allylation of aldehydes.

3.5 Other Applications

The intramolecular hydroamination of alkenes constitutes a powerful method for the synthesis of azacycles.[68]Livinghouse reported the structure of the first low-coordinate yttrium–bis(thiophosphinic amidate) complex determined by X-ray diffraction.[69] The complex had a distorted square-pyramidal geometry with the bis(trimethylsilyl)amido ligand at the apex. In addition, the use of this complex, as well as related chelates of representative group 3 and lanthanide metals (yttrium, neodymium, and dysprosium),[69] or neutral zirconium(IV) metal[70] for the efficient catalysis of intramolecular alkene hydroaminations was studied in his group (Schemes 22 and 23). The corresponding cyclized amination products were obtained in good yields using these thiophosphoramide ligands. These ligands have pseudotetrahedral bonding geometry at phosphorus for increased steric shielding of the complexed metal relative to the planar central carbon present



Scheme 22



Scheme 23

in the more familiar amidinato[71] group and a diminished predisposition toward aggregate formation with oxophilic metals by virtue of terminal sulfur (versus oxygen) substitution. These advantages mean that there is higher efficient catalytic activity than when amine ligands are used in the intramolecular alkene hydroaminations. In addition, up to 89% ee has been achieved in the asymmetric intramolecular hydroamination of alkenes.[72]

4 Conclusion

For a long time, asymmetric carbon–carbon bond-formation reactions have been useful methods in modern organic synthesis. In the past two decades, various phosphorus, nitrogen and sulfur ligands have been widely used in asymmetric addition reactions. However, thiophosphoramide and thioamide ligands have had few applications in asymmetric catalytic reactions. Our group has made much effort in the synthesis of thiophosphoramide and thioamide ligands, which are easily available, quite stable, recoverable and reusable. We have also looked at their use in the conjugate addition of organozinc reagents to enones, addition of diethylzinc to imines, addition of diethylzinc to aldehydes, and addition of allylic stannane to aldehydes. As these methods are now available, they can be expected to be applied in the asymmetric synthesis of natural products in the near future.

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